

Industry 4.0

With regards to Industry 4.0 circa 2020: *“The perceived level of autonomy of the applications is still generally quite low.”*

There seem to be some over generalization on what the Industrial sector is. AI could potentially solve a HUGE range of things, from logistics: fleet management, inventory, shipments, self-driving vehicles, warehouse management, to Manufacture: CAD/BIM modeling, factory assembly, take offs and quantities, safety audits, testing and QA, to collection of raw materials: location of resources, optimization of process methods, greener manufacturing, etc..

“from defect classification in quality control, to fault detection in predictive maintenance, ergonomics and human assistance in manual tasks.”

However many of these things are not specific to the Industrial sector.

Note that much of this is Lab-based studies. Many with synthetic data or using SMOTE (Synthetic Minority Over-sampling Technique). But notes that the results of this are “underwhelming”.

semi-supervised learning - for data-sets too large to label

“making data quality a critical factor for the industrial success of these solutions”

Accessible Representative Contextual Intrinsic

Peres, R.S., Jia, X., Lee, J., Sun, K., Colombo, A.W. and Barata, J. (2020) ‘Industrial Artificial Intelligence in Industry 4.0 - Systematic Review, Challenges and Outlook’, *IEEE Access*, 8, pp. 220121–220139. Available at: <https://doi.org/10.1109/ACCESS.2020.3042874>

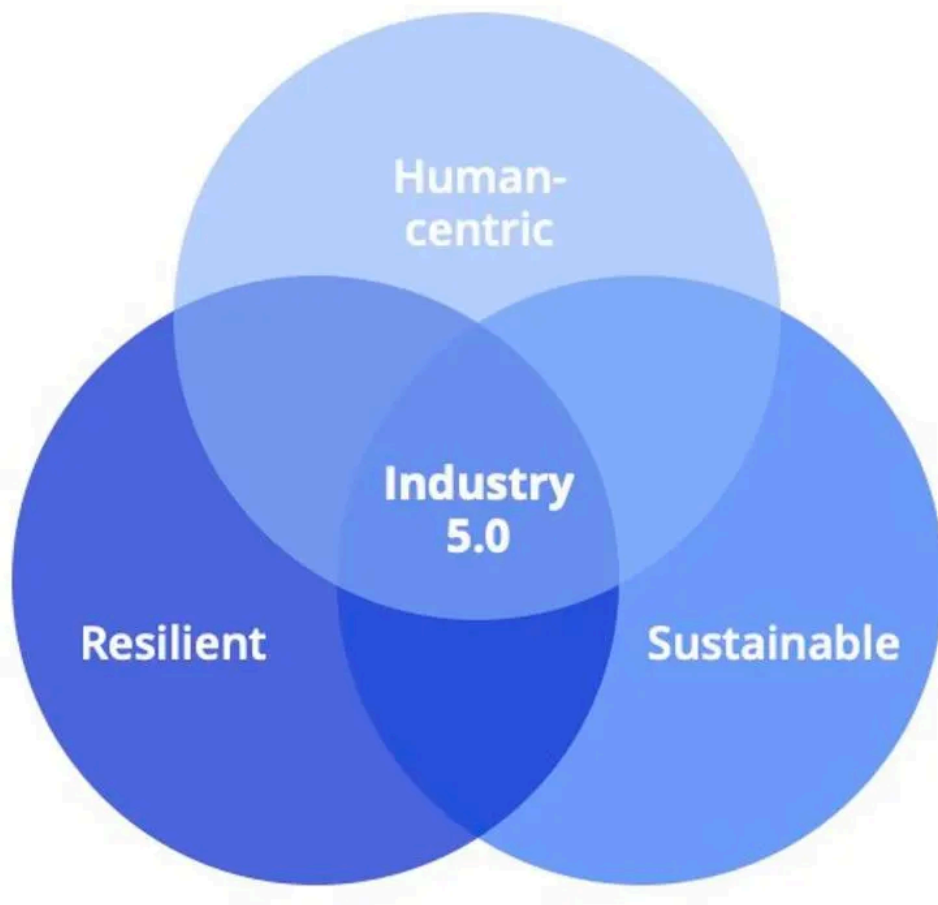
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Whilst intentionally good, the concept seem somewhat idealized. It may not translate into the limitations, constraints and market-driven pressures of real-world business environments.

“specifically putting research and innovation at the service of the transition to a sustainable, human-centric and resilient European industry.” (European Commission, no date)

Whilst previous revolutions have been lead by technology and innovation, this seems to want to harness the technology to “do good”. Although the European union may defend the idea that sustainable business practices are needed to be prosperous in the modern world, they fail to back these up with concrete evidence to support this.

Kraaijenbrink, J., (2022) notes that modern business tend towards a focus on efficiency and profit.



The Three Pillars of Industry 5.0

Based on the EU publication "Industry 5.0 Human-centric, sustainable and resilient"

European Commission (no date) *Industry 5.0*. Available at:

https://research-and-innovation.ec.europa.eu/research-area/industrial-research-and-innovation/industry-50_en (Accessed: 7 May 2026).

Kraaijenbrink, J. (2022) *What Is Industry 5.0 And How It Will Radically Change Your Business Strategy*. *Forbes*, 24 May. Available at:

<https://www.forbes.com/sites/jeroenkraaijenbrink/2022/05/24/what-is-industry-50-and-how-it-will-radically-change-your-business-strategy/> (Accessed: 7 May 2026).